

Ayush Kumar Singh

Java Backend Engineer | Spring Boot | REST APIs | Microservices | Production Ownership
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ayushsingh45portfolio.vercel.app | Kolkata, India

Summary

Java Backend Engineer with 2+ years building production Spring Boot services for an insurance platform used by 500K+ policyholders. Own backend delivery end-to-end – REST API design, Spring Data JPA/Hibernate data layer, and UAT-to-production release – across Oracle and PostgreSQL. Reduced batch query execution time by 70% and raised unit test coverage from 4% to 70%, cutting regression risk across releases.

Professional Experience

Associate Software Engineer – Java Backend
Indus Net Technologies, Kolkata

Jun 2024 – Present
Full-Time

- Shipped the Self & Spouse Insurance Policy Module for the SahajBeema platform (with Niva Bupa) end-to-end, by designing and owning independently-deployable Spring Boot services, REST APIs, and Hibernate/JPA data models for insurer-specific integration workflows from build through UAT.
- Delivered a zero-downtime Oracle data migration affecting 5 lakh+ user records, by writing and validating production migration queries for Niva Bupa's master configuration update.
- Prevented a production outage during a third-party insurer (Bajaj) domain migration, by implementing exception-handled fallback logic for their API endpoints ahead of the cutover.
- Cut batch-processing query execution time by 70% and eliminated recurring deadlocks, by rewriting high-volume Oracle queries using SQL MERGE statements and query-plan optimization.
- Reduced maintainability issues from 370 to 165 and raised unit test coverage from 4% to 70%, by leading a SonarQube clean-code initiative using JUnit 5 and Mockito integrated into the CI pipeline.
- Improved eKYC verification reliability for HDFC's secondary policy flow, by migrating the third-party API integration from Karza to Surepass with backward-compatible error handling.
- Streamlined the insurance admin portal's usability and access control, by designing pagination, filtering, role-based authentication/authorization, and SQL-driven reporting across insurer-wise policy and claims dashboards.
- Reduced release friction for the QA team, by owning UAT deployments across both the React frontend and backend services using PuTTY and WinSCP for Linux server access and build deployment.
- Improved production reliability, by installing and maintaining scheduled cronjobs, debugging failing API endpoints, and proactively detecting and triaging production issues via server log analysis.

Projects

Task Management Backend System

github.com/ayush45-code/task-management

Java 17, Spring Boot 3, Spring Security, PostgreSQL, JWT, Spring Data JPA, Hibernate

- Architected a task management-lifecycle backend on Java 17 and Spring Boot 3, integrating both JWT and OAuth services for Login.
- Enforced role-based access control on every endpoint, by implementing JWT-based authentication and authorization with Spring Security.
- Designed REST APIs and the Hibernate/JPA persistence layer over PostgreSQL, including indexed queries for task lookups and assignment history.

Personal Portfolio

ayushsingh45portfolio.vercel.app

React, JavaScript, HTML5/CSS3, Vercel (CI/CD)

- Kept the live site always in sync with the latest code, by setting up continuous deployment on Vercel triggered from GitHub pushes.

Technical Skills

Core Backend: Java, Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, API Integration

Database & Performance: Oracle Database, PostgreSQL, MySQL, Query Optimization

Testing & Quality: JUnit 5, Mockito, SonarQube, Unit Testing

Practices: Agile/Scrum, SOLID Principles, OOP, Design Patterns, Clean Code, Production Support/Debugging

Supporting Tools: Git, GitHub, Docker, AWS (EC2, S3, IAM), Postman, Linux, CI/CD, n8n

Frontend (Supporting): ReactJS, JavaScript, HTML5/CSS3

Certifications

AWS Academy Graduate – Cloud Foundations

Nov 2023

AWS Academy Graduate – Introduction to Cloud, Semester 1

Oct 2023

AWS Academy Graduate – Introduction to Cloud, Semester 2

Oct 2023

Education

Bachelor of Technology in Computer Science & Engineering

Adamas University, Kolkata

2020 – 2024

CGPA: 8.45 / 10.0